

CANCER

Microenvironment actuated CAR T cells improve solid tumor efficacy without toxicity

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A major limiting factor in the success of chimeric antigen receptor (CAR) T cell therapy for the treatment of solid tumors is targeting tumor antigens also found on normal tissues. CAR T cells against GD2 induced rapid, fatal neurotoxicity because of CAR recognition of GD2⁺ normal mouse brain tissue. To improve the selectivity of the CAR T cell, we engineered a synthetic Notch receptor that selectively expresses the CAR upon binding to P-selectin, a cell adhesion protein overexpressed in tumor neovasculature. These tumor microenvironment actuated T (MEAT) cells ameliorated T cell infiltration in the brain, preventing fatal neurotoxicity while maintaining antitumor efficacy. We found that conditional CAR expression improved the persistence of tumor-infiltrating lymphocytes because of enhanced metabolic fitness of MEAT cells and the infusion of a less differentiated product. This approach increases the repertoire of targetable solid tumor antigens by restricting CAR expression and subsequent killing to cancer cells only and provides a proof-of-concept model for other targets.

INTRODUCTION

Chimeric antigen receptor (CAR) T cell therapy is an adoptive cell therapy that relies on T cells engineered to express an antibody-based receptor to redirect the effector T cells to eliminate cancer cells (1). CAR T cell therapy has proven remarkably effective in the treatment of B cell neoplasms, demonstrating high response rates (70 to 90%) in clinical trials and resulting in the Food and Drug Administration approval of six CAR T cell therapies (2–4). However, a lack of cell surface tumor-specific antigens has played a major role in limiting the clinical application of CAR T cells to the treatment of solid tumors (5, 6). Although CD19- and B cell maturation antigen-targeted CAR T cells are B cell lineage restricted, their side effects are not lethal and can be managed by infusion of immunoglobulins (7, 8). Conversely, targeting antigens found on solid tumors can have potentially fatal toxicities because of target antigen expression on essential normal tissues (5).

The disialoganglioside GD2 is an appealing target for several cancers, especially neuroectodermally derived cancers, such as neuroblastoma. However, therapeutic antibodies targeting GD2, such as dinutuximab, have short half-lives in patients relative to CAR T cells and have resulted in severe peripheral nerve toxicities because of recognition of GD2⁺ nerve cells (9–11). A high-affinity single-chain variable fragment (scFv) against GD2 (E101K) in CAR T cells showed improved preclinical efficacy compared to the lower affinity wild-type (WT) scFv (12). Further development of the E101K CAR for clinical use has been complicated, however, because of fatal encephalitis in mice resulting from recognition of normal GD2⁺ cells in the brain. Although clinical trials of GD2 CAR T cells with the WT scFv demonstrated minimal neurotoxicity in humans, only 33% of patients demonstrated a complete response (13). Use of the more efficacious

affinity-enhanced scFv may greatly improve patient outcomes by increasing the sensitivity of the CAR to target cells expressing low levels of GD2, if on-target neurotoxicity could be circumvented.

CAR T cells against HER2 and ROR1 further highlight this issue of “on-target, off-tumor” toxicity, as demonstrated by fatal respiratory toxicity in patients and bone marrow failure in preclinical models, respectively (14, 15). Therefore, systemic targeting of tumor antigens also found on normal tissue remains an important hurdle in the translation of CAR T cells to safe and effective solid tumor treatment.

One approach to ameliorate these toxicities is to restrict CAR expression to the tumor site, thereby sparing healthy tissue (16). A method using synthetic Notch (synNotch) actuator receptors was recently developed by W. Lim’s lab to allow for such conditional CAR expression (17–19). After synNotch receptor binding to its target ligand, its transcription factor initiates transcription of a transgene of interest, such as a CAR. While the utility of the synNotch system has been demonstrated by several groups, there is a need for tumor-specific and clinically relevant actuator ligands to prevent the on-target toxicity of CARs against solid tumor targets (15, 20–24). Specifically, targeting normal ligands within the tumor microenvironment (TME) by the synNotch actuator would limit actuator signal escape because of the reduced plasticity of normal vascular cells, minimizing potential for mutation or down-regulation of the actuator target.

In this work, we identified P-selectin as a TME-specific antigenic actuator of synNotch circuits. P-selectin is an inflammatory cell adhesion molecule found on activated endothelial cells that plays a role in T cell trafficking and migration in response to injury (25–27). Selectins have also been shown to facilitate cancer metastasis, and elevated levels of soluble P-selectin are associated with poor patient prognosis (28–31). Recent work in preclinical murine models demonstrated antitumor efficacy using anti-P-selectin (αP-selectin) antibodies or targeted nanoparticles (27, 32–35). Furthermore, P-selectin is highly expressed in a wide range of solid cancer TMEs, including lymphoma, ovarian, breast, lung, and others, with limited expression in adjacent normal tissue (33). P-selectin is therefore an attractive actuation ligand for a diverse repertoire of CAR T cells across a spectrum of solid tumors by harnessing T cell biology to specifically home to the TME before CAR T cell cytotoxicity is initiated.

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We developed tumor microenvironment actuated T cells (MEAT cells) using an α P-selectin synNotch to restrict expression of the CAR to malignant tissue. We designed MEAT cells gating both the CD19 CAR, as a clinically validated CAR and proof of concept, and the high-affinity, neurotoxic GD2 CAR to evaluate specificity and antitumor efficacy of P-selectin-gated systems in murine models of neuroblastoma. MEAT cells demonstrated P-selectin-specific actuation and CAR expression, activation, and killing in vitro. In comparison to conventional GD2 CAR T cells that were rapidly lethal to mice because of brain infiltration, a single infusion of P-selectin-gated GD2 CAR T cells in mice bearing neuroblastoma tumors demonstrated antitumor efficacy without detectable neurotoxicity or T cell infiltration into the brain. In a second model, we found that gating the CD19 CAR improved antitumor efficacy by improving metabolic fitness and maintaining a less differentiated phenotype, as compared to constitutive

CD19 CAR T cells. Thus, MEAT cells may expand the therapeutic index of CARs by preventing fatal toxicity and improving functionality. This modular system has the potential to expand to other solid tumor targets, thus providing a general strategy to improve the safety and efficacy of CAR T cells for solid tumors in the clinic.

RESULTS

Conventional GD2 CAR T cells cause fatal neurotoxicity in immunodeficient mice

We first sought to assess the toxicity of the high-affinity GD2 CAR (E101K) and to elucidate the kinetics, biodistribution, and mechanism of neurotoxicity. We cloned the GD2 CAR into a pHR lentiviral expression vector and transduced primary human T cells from healthy donors (Fig. 1A). NSG mice were intravenously engrafted

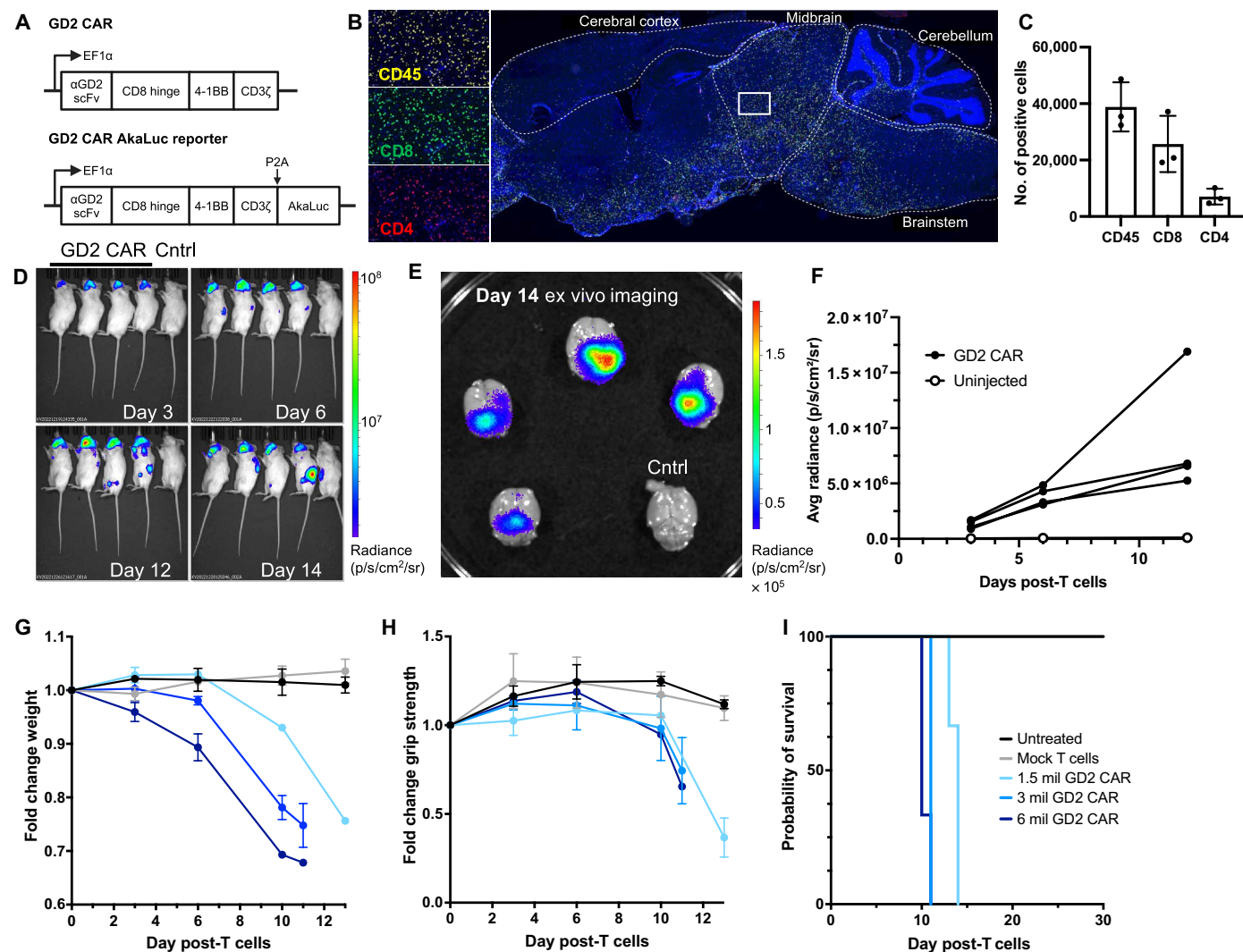


Fig. 1. GD2 CAR T cells cause fatal neurotoxicity in immunodeficient mice. (A) Design of lentiviral α GD2CAR construct and α GD2CAR AkaLuc reporter. (B) Representative image of CD45 histological staining of NSG mouse brains treated with primary human α GD2CAR T cells. Left-hand panels show zoom of white box region of interest. Comparative analysis of CD4- and CD8-positive brain-infiltrating populations of primary human GD2CAR T cells. (C) Quantification of brain-infiltrating cells ($n = 3$ mice). (D) Time course of GD2 CAR T cell expansion via BLI of AkaLuc expressing primary T cells ($n = 4$ mice) or control-injected mice (Cntrl, mock T cell not expressing AkaLuc). All mice are in indicated order as in the top-left panel. (E) Ex vivo imaging of mouse brains on day 14 end point by BLI. (F) Quantification of brain-specific CAR T cell expansion. Quantification of mouse weight (G), grip strength (H), and overall survival (I) for a dose titration of GD2 CAR T cells ($n = 3$ mice per group). mil, million. Data are means with individual data points (C) or means $\pm$ SD [(F) to (I)].

with GD2 CAR T cells, and their brains were harvested after 10 days to evaluate T cell infiltration in the brain. Immunofluorescence (IF) staining of CD45⁺, CD4⁺, and CD8⁺ cells in the brain showed significant T cell infiltration with CD8⁺ cells being the predominant hematopoietic cell subset (3:1 ratio of CD8⁺ to CD4⁺ cells), suggesting preferential expansion of CD8⁺ cells, as observed by others (Fig. 1, B and C) (36). T cells were found in brain regions known to express GD2, including the midbrain and brainstem, and were absent in brain regions with low or no GD2, such as the cerebral cortex (12). We observed similar neurological toxicities and T cell brain infiltration in mice preconditioned with cyclophosphamide and when injecting CAR T cells via different intravenous routes (fig. S1).

Mice were then intravenously injected with GD2 CAR T cells expressing AkaLuc as a reporter and tracked via bioluminescence imaging (BLI) to determine kinetics and biodistribution of CAR expansion (Fig. 1A). GD2 CAR T cells localized to the brain by day 3 and continued to expand through the end point on day 14 (Fig. 1, D to F). Brain-specific localization was confirmed by ex vivo imaging of the whole brain after injection of the AkaLuc substrate AkaLumine-HCl. CAR T cells appeared to also expand systemically, as shown by a signal in the spleen on day 6 and lymph nodes, thymus, and lungs on day 12 (fig. S2).

To determine systemic toxicity, healthy NSG mice were engrafted with GD2 CAR T cells at varying doses (1.5×10^6 , 3×10^6 , or 6×10^6). Mice were weighed periodically and euthanized upon 20% weight loss, as per Institutional Animal Care and Use Committee (IACUC) protocols. We found a dose-response relationship in toxicity with mice rapidly losing weight after injection and demonstrating impaired grip strength, an indicator of neurotoxicity, relative to control mock T cell-treated mice (Fig. 1, G and H). Control mock T cells were spinoculated using the same protocol as transduced CAR T cells but without the addition of virus. All GD2 CAR T cell doses led to sacrifice within 2 weeks because of severe weight loss and onset of clinical neurological distress (circling behavior, hunched posture, and hindlimb paralysis) (Fig. 1I). Control mice injected with CD19 CAR T cells did not demonstrate weight loss or significant T cell brain infiltration (fig. S3). These data are consistent with previous reports that high-affinity GD2 CAR T cells infiltrate the brain because of on-target reactivity, leading to fatal encephalitis in mice.

P-selectin is an antigenic actuator in the TME

We next sought to identify a suitable target in the TME that is highly expressed in a wide range of cancer types to use as a synNotch actuator. Actuator targets currently under development use tumor-associated antigens to gate synNotch circuits, allowing potential escape from actuation because of antigenic heterogeneity or loss resulting from the plasticity of cancer cells (21–24). We investigated the cell adhesion molecule P-selectin because of its high expression in solid tumor vasculature, minimal expression in normal tissue, and biological role in cancer cell metastasis and T cell homing to sites of inflammation. Histological staining of P-selectin in samples from 27 patients with primary neuroblastoma showed high expression of P-selectin on CD31⁺ endothelial cells (Fig. 2A).

In a subset of patients, P-selectin was also observed at high levels on tumor or other stromal cells (fig. S4). There was minimal quantitative expression of P-selectin on normal peripheral nerve and brain tissues relative to tumor, minimizing potential off-tumor actuation and toxicity in normal GD2⁺ tissue (Fig. 2B and fig. S5). These data agree with previous work reporting minimal detection of P-selectin

in the midbrain, periventricular, hippocampal, and white matter regions in normal brain tissue, with only the choroid plexus region of the central nervous system expressing detectable levels (27, 37–40).

Primary T cells can be engineered to gate CARs using an α P-selectin synNotch

We next engineered MEAT cells to express an anti-human P-selectin (α hSELP) synNotch to control CAR expression. We hypothesize that MEAT cells expressing an α hSELP synNotch actuate in the presence of human P-selectin to transcribe, translate, and express the CAR on the surface and selectively kill antigen-positive cancer cells in the local environment (Fig. 2C). The scFv for crizanlizumab, a Food and Drug Administration–approved α P-selectin antibody used to treat patients with sickle cell anemia, was used to generate a P-selectin–specific synNotch (41).

We cloned the α hSELP synNotch into a single two-part vector with (i) a constitutive response element containing the α P-selectin synNotch and (ii) an actuation element beneath the conditional TRE (tetracycline response element) promoter with either the CD19 or GD2 CAR (Fig. 2D). Primary human T cells were transduced with P-selectin–sensing synNotch constructs, and synNotch and CAR surface expression was validated by flow cytometry using anti-myc tag and anti-idiotypic antibodies, respectively (Fig. 2E and fig. S6A). We then cultured Jurkat MEAT cells with plate-bound human P-selectin and assessed actuation after 24 hours. P-selectin–specific CAR expression was observed with >75% of synNotch⁺ cells actuating and presenting CD19 CAR on the surface (Fig. 2F and fig. S6B).

α P-selectin synNotch cells are specific, sensitive, and cytotoxic in vitro

We next sought to characterize MEAT cell actuation, activation, and killing kinetics. We found that primary MEAT cells only expressed CD19 CAR in the presence of plate-bound P-selectin (Fig. 3A). Soluble P-selectin did not induce CAR expression, likely because of a lack of mechanical force necessary to expose protease cleavage sites. This observation is critical to eliminate off-tumor CAR expression in physiological environments where soluble P-selectin might circulate in the plasma to promote inflammation and coagulation. CAR surface expression increased over 24 hours after stimulation with target cells (Fig. 3B) and decreased quickly over time in the absence of the target, suggesting minimal off-tumor killing should T cells migrate to the brain after engaging with P-selectin in the TME (fig. S6C).

We then established a T cell coculture with GD2⁺ T98G glioblastoma target cells, either expressing P-selectin or not expressing P-selectin, to monitor CAR T cell activation kinetics. Constitutive GD2 CAR T cells activated rapidly within the first 24 hours and then decreased in activation over the following 48 hours, as measured by CD69 surface expression (Fig. 3C and fig. S7). Conversely, α hSELP synNotch $\rightarrow$ GD2 CAR T cells showed delayed activation with peak CD69 expression at 48 hours. There was no CD69 detected on synNotch CAR T cells when cocultured with P-selectin–negative target cells, demonstrating actuator-specific activation (Fig. 3C, dashed line).

Next, we established a live-cell coculture with effector cells, Lenti-X actuator cells (stably expressing P-selectin), and SK-N-BE(2) neuroblastoma target cells (CD19⁺ or GD2⁺) to understand the kinetics of target cell killing (Fig. 3D). Specific and dose-dependent killing was observed for both α hSELP synNotch $\rightarrow$ CD19 CAR and

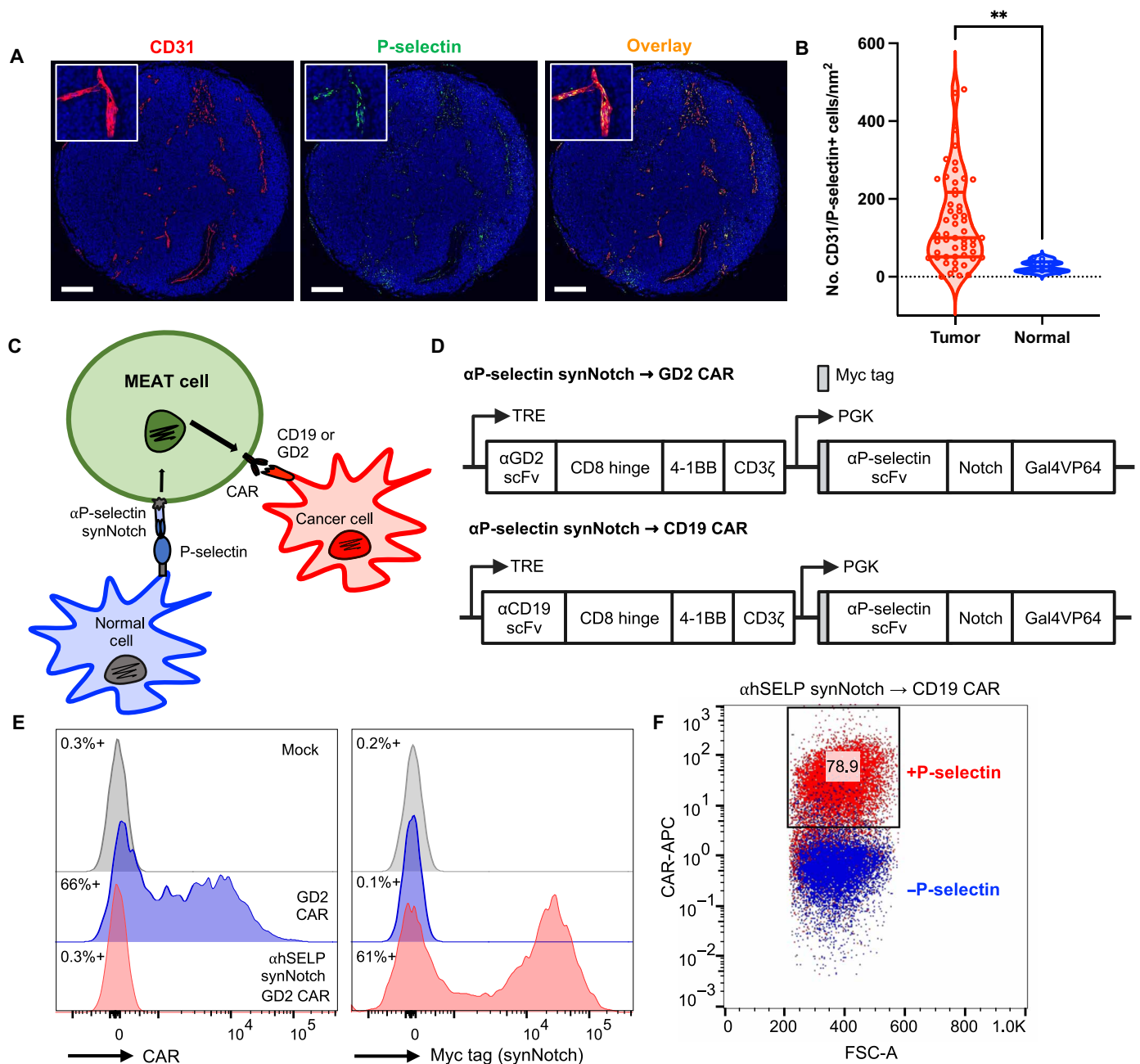


Fig. 2. P-selectin is an actuator target in the TME. (A) Representative IF stain of human neuroblastoma patient tissue for P-selectin (green), CD31 (red), and their colocalization (orange). Scale bars, 200 μm . (B) Quantification of P-selectin/CD31 expression in human neuroblastoma and normal peripheral nerve tissue via tissue microarray ($n = 27$ tumor, $n = 9$ normal). (C) MEAT cell system. MEAT cells contain an anti-human P-selectin synNotch, which induces expression of either CD19 or GD2 CAR upon engagement to P-selectin. (D) $\alpha\text{hSELP synNotch}$ constructs actuating either the GD2 CAR or CD19 CAR. (E) Transduction efficiency of $\alpha\text{hSELP synNotch} \rightarrow \text{GD2 CAR}$ MEAT primary T cells. (F) Jurkat MEAT cell actuation by plate-bound recombinant human P-selectin. FSC-A, forward scatter area. Data represented by the violin plot are individual data points, and statistics were calculated using unpaired t test (B). ** $P \leq 0.01$.

constitutive CD19 CAR T cells at varying effector-to-actuator-to-target cell ratios (E:A:T) (Fig. 3, E and F). An approximate 15- to 24-hour delay in 50% killing was observed for synNotch CAR T cells because of de novo biosynthesis of the CAR after engaging with P-selectin, as compared to the constitutive CAR T cells. These data support the activation kinetics previously described, which also

show a 24-hour delay in activation. $\alpha\text{hSELP synNotch} \rightarrow \text{GD2 CAR}$ T cells selectively killed in the presence of both actuator and target ligands, as shown by killing of SK-N-BE(2) (CD19 $^{+}$ mCherry $^{+}$) target cells after a 72-hour coculture, and spared antigen-negative GFP $^{+}$ “bystander” cells (Fig. 3G). Live-cell coculture experiments were repeated in both *cis* (P-selectin and CAR target on the same

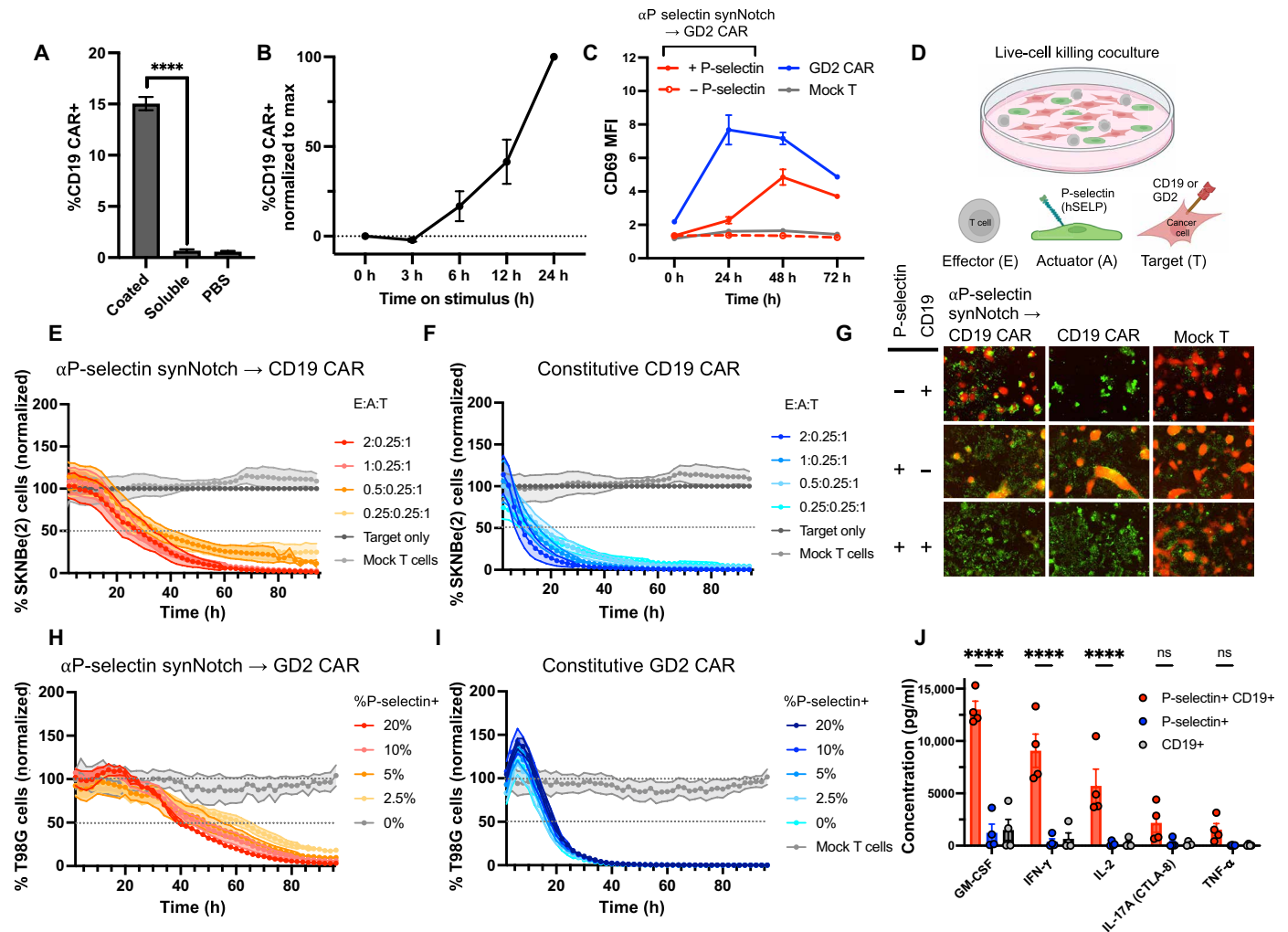


Fig. 3. MEAT cells are specific and cytotoxic. (A) Actuation of primary αhSELN synNotch → CD19 CAR MEAT cells by soluble and plate-bound recombinant human P-selectin ($n = 3$ technical replicates). (B) On kinetics of CD19 CAR in αhSELN synNotch → CD19 CAR MEAT primary T cells incubated with recombinant human P-selectin ($n = 3$ technical replicates). h, hours. (C) Time course of antigen-specific activation in αhSELN synNotch → GD2 CAR MEAT cells cocultured with GD2+ target cells as measured by CD69 expression. (D) Live-cell killing coculture schematic of effector (E), actuator (A), and target (T) cells. Live-cell killing of (E) αhSELN synNotch → CD19 CAR MEAT cells and (F) CD19 CAR T cells at varying effector cell concentrations. (G) Fluorescence imaging of actuator (green) and target (red) cells cocultured with either synNotch CD19 CAR, CD19 CAR, or mock T cells (unlabeled) for 72 hours (representative images of three technical triplicates). Live-cell killing of (H) αhSELN synNotch → GD2 CAR MEAT cells and (I) GD2 CAR T cells at varying actuator cell concentrations. The gray dotted line represents 50% killing of target cells. (J) Top five cytokines secreted after 72-hour coculture of effector cells with target cells expressing both the actuator and antigen (P-selectin+ and CD19+) or only one present (P-selectin+ or CD19+) as assayed by the Luminex 25-plex human cytokine panel. Data are shown as means $\pm$ SD, and statistics were calculated using one-way ANOVA (A) or two-way ANOVA with Tukey's post hoc test (J) where ns > 0.05 , **** $P \leq 0.001$.

cell) and *trans* (P-selectin and CAR target on different cells) using peripheral blood mononuclear cells from several healthy donors, and similar trends were observed (figs. S8 and S9).

We then titrated the percentage of P-selectin+ cells to probe the sensitivity of our synNotch to actuate in response to decreasing abundance of the actuator ligand. Although the time to 50% killing was delayed with decreasing frequency of P-selectin+ cells, αhSELN synNotch → GD2 CAR T cells were able to completely eradicate target cells with as low as 2.5% P-selectin+ actuator cells present (Fig. 3H and fig. S10). Conversely, constitutively active GD2 CAR T cells showed no dependence on decreasing concentration of P-selectin (Fig. 3I). Furthermore, we observed specific cytokine secretion dependent on the presence of both actuator and target ligands,

as measured by Luminex using a 25-plex human cytokine array. The five most highly secreted cytokines included GM-CSF (granulocyte-macrophage colony-stimulating factor), IFN- γ (interferon- γ), IL-2 (interleukin-2), CTLA-8 (cytotoxic T lymphocyte-associated antigen 8), and TNF- α (tumor necrosis factor- α), all inflammatory cytokines associated with T cell activation and cytotoxic function (Fig. 3J). Incubation of synNotch T cells with target cells expressing either an actuator or target ligand failed to produce an inflammatory response. Similar trends were observed in additional cytokines profiled, including IL-22, TNF- β , IL-13, IL-10, and IL-1RA (IL-1 receptor antagonist) (fig. S11). These data demonstrate that αhSELN synNotch T cells maintain the cytotoxicity of the gated CAR specifically in the presence of P-selectin and that only a small fraction of

the tumor would need to be P-selectin⁺ for the cells to activate and kill the cancer cells.

Conditional CAR expression improves T cell fitness and functionality

Next, we investigated the phenotypic and functional consequences of conditional CAR expression on T cell fitness. We found that resting (unstimulated) synNotch CAR T cells expressed significantly fewer exhaustion markers (Tim3, Lag3, and PD1) and maintained a more naïve-like phenotype (CD62L⁺, CD45RA⁺, and CD45RO⁻) 7 days after manufacture, as compared to constitutive CAR T cells

(Fig. 4, A and B, and fig. S12). These data suggest that eliminating CAR expression during manufacturing via controlled synNotch gating may prevent premature differentiation and exhaustion in preinfusion products.

Despite these phenotypic differences, there was no significant change in cytotoxicity when unstimulated effector cells were cocultured with target cells (Fig. 4C). To understand CAR T cell function via serial killing of target cells, we set up a chronic activation assay where we stimulated effector cells with target cells for 15 days, periodically refreshing target cells for a total of four stimulations. After 15 days of co-incubation, we found an increase in exhaustion

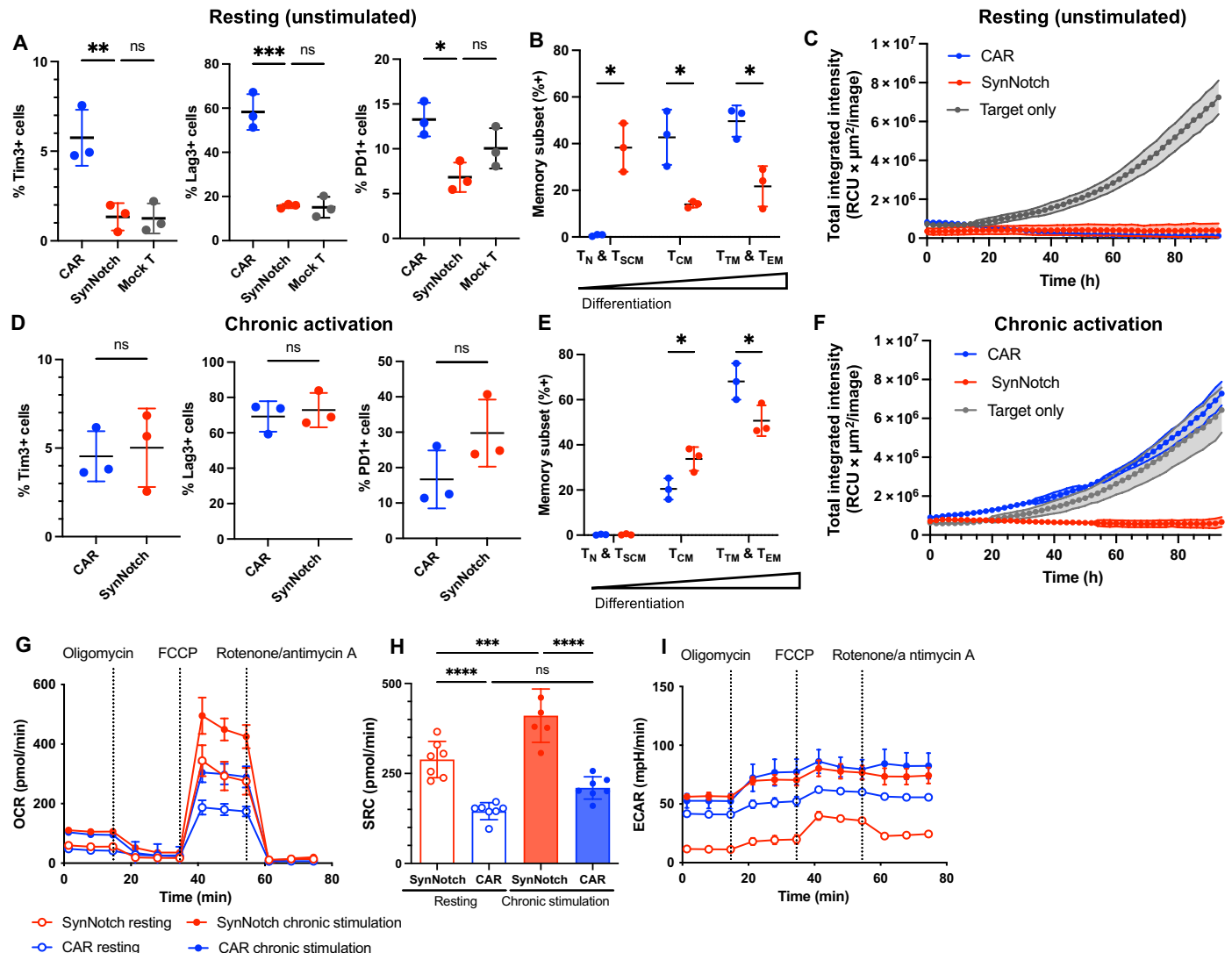


Fig. 4. MEAT cells are less differentiated and more metabolically fit. (A to C) Resting (unstimulated) CD19 CAR, synNotch, or mockT cells assayed for surface expression of (A) Tim3, Lag3, and PD1 and (B) memory phenotype by flow cytometry. (C) SK-N-BE(2) target cell killing of unstimulated effector cells (E:T of 1:1). (D to F) CART cells were chronically activated for 15 days by coculture with 20% actuator (P-selectin⁺) and 80% target (CD19⁺) cells and with fresh actuator and target cells added on days 4, 8, and 12. Chronically activated CAR T cell (D) Tim3, Lag3, and PD1 surface expression and (E) memory phenotype were assessed by flow cytometry. (F) SK-N-BE(2) target cell killing of chronically activated effector cells (E:T of 1:1). [(G) to (I)] Metabolic analysis of effector cells following 15-day stimulation as measured by Seahorse mitochondrial stress test. (G) OCR, (H) SRC, and (I) ECAR. Flow data are shown as means ± SEM (*n* = 3 biological replicates). Cytotoxicity and Seahorse data are shown as means ± SEM (*n* = 3 technical replicates, *n* = 2 biological replicates). T_N (naïve), T_{SCM} (stem cell memory), T_{CM} (central memory), T_{TM} (transitional memory), and T_{EM} (effector memory). Data are shown as means ± SD, and statistics were calculated using one-way ANOVA with Dunnett's post hoc test [(A) and (B)], unpaired *t* test [(B) and (E)], or one-way ANOVA with Tukey's post hoc test to correct for multiple comparisons (H) where *ns* > 0.05, **P* ≤ 0.05, ***P* ≤ 0.01, ****P* ≤ 0.001.

marker expression on synNotch T cells to reach the same level as constitutive CAR T cells (Fig. 4D). SynNotch CAR T cells retained a less differentiated phenotype, as observed in the resting MEAT cells (Fig. 4E). We then performed a killing assay using chronically activated effector cells and found that synNotch CAR T cells maintained cytotoxic function, while constitutive CAR killing was completely abolished (Fig. 4F). These data suggest that synNotch CAR T cells may enhance efficacy *in vivo* because of better persistence, less dysfunction, and longer-term memory. Furthermore, exhaustion marker expression is not fully predictive of T cell function, and other factors may influence cytotoxicity after chronic antigen stimulation.

T cell fate is also strongly influenced by metabolic preference to undergo glycolysis or oxidative phosphorylation. After antigen engagement, naïve T cells activate, proliferate, and differentiate into effector cells, favoring the less efficient aerobic glycolysis to support rapid cell growth. Cytotoxic effector cells then differentiate into memory T cells after antigen clearance, favoring oxidative metabolism. The ability of T cell mitochondria to switch freely between these various metabolic processes, termed “metabolic flexibility,” is critical to maintain energy and homeostasis in response to disease and is a strong determinant of patient outcome after CAR T cell treatment (42, 43).

Given that MEAT cells maintain a less differentiated phenotype, and T cell differentiation is strongly interconnected to metabolism, we next investigated the impact of synNotch gating T cell bioenergetics. We used the Seahorse mitochondrial stress test to probe changes in oxygen consumption rate (OCR; a measure of oxidative phosphorylation) and extracellular acidification rate (ECAR; a measure of glycolysis) in resting (unstimulated), acutely (48 hours) activated, and chronically (15 days) activated effector cells. CAR T cells were assayed for mitochondrial fitness through serial additions of oligomycin (complex V adenosine 5'-triphosphate synthase inhibitor), carbonyl cyanide *p*-trifluoromethoxyphenylhydrazone (FCCP; uncoupler of oxidative phosphorylation), and rotenone and antimycin A (complex I and III inhibitors).

Resting MEAT cells exhibited a significantly higher maximal OCR after FCCP addition, as compared to resting constitutive CAR T cells (Fig. 4G). Similarly, chronically activated synNotch T cells had a significantly higher maximal OCR than chronically activated constitutive CAR T cells, which translated to a higher spare respiratory capacity (SRC) in synNotch T cells, demonstrating their higher metabolic fitness under both resting and chronically stimulated T cell conditions (Fig. 4H). Similar trends were observed in acutely activated effector cells (fig. S13). We simultaneously assessed changes in ECAR, a measurement of bulk acidification because of changes in glycolysis and/or oxidative phosphorylation. At baseline, resting synNotch T cells had the lowest ECAR, indicating minimal bioenergetic activity (Fig. 4I). These data support the less differentiated memory phenotype of resting and chronically stimulated MEAT cells and suggest that conditional CAR expression may promote more favorable bioenergetics, improving cell metabolic fitness and function.

α P-selectin synNotch GD2 CAR T cells are nontoxic in healthy mice

To test whether our gated cells were safe in mice, we engrafted healthy NSG mice with either constitutive GD2 CAR T cells, α hSCLP synNotch $\rightarrow$ GD2 CAR T cells, or control mock T cells. GD2 CAR T cell-treated mice lost >20% body weight by day 12 after T cell

injection, whereas synNotch CAR T cell-treated mice had no significant change in weight (Fig. 5A). SynNotch CAR-treated mice showed no signs of neurotoxicity, whereas GD2 CAR-treated mice exhibited circling behavior, hunched posture, and hindlimb paralysis, as previously described. Mice were euthanized on days 1, 5, 9, and 12 to capture the kinetics of CAR expansion in the brain (Fig. 5, B and C). GD2 CAR T cells were undetectable in the brain through day 9, supporting the minimal change in weight during that period. By day 12, there was a significant spike in CD8⁺ T cells infiltrating the brain in GD2 CAR T cell-treated mice, whereas synNotch and mock T cell cohorts had no detectable infiltration.

We also observed enlarged spleens in GD2 CAR-treated mice through day 9 with a return to baseline at the end point and no change in weight for synNotch or mock T cell cohorts. Despite the decrease in spleen weight by day 12, there was still a significantly higher proportion of both human CD3⁺ T cells and GD2 CAR T cells by *ex vivo* flow cytometry relative to the controls (Fig. 5, E and F). There were also no detectable GD2 CAR T cells in the brain in synNotch-treated mice by flow cytometry, supporting the results from IF staining (Fig. 5G). Together, these data demonstrate that α hSCLP synNotch $\rightarrow$ GD2 CARs prevent on-target, off-tumor neurotoxicity in healthy mice.

α P-selectin synNotch GD2 CAR T cells ameliorate neurotoxicity in solid tumor models

Assessing P-selectin-specific killing *in vivo* would require a synNotch-gated CAR T cell that recognizes murine P-selectin in TME blood vessels and GD2 on the human xenograft cells. Although extensive efforts were made to generate an anti-mouse P-selectin synNotch, the four published scFv sequences we were able to obtain were ineffective (32, 41, 44–47).

Therefore, we constructed a xenograft model that would mimic the *in vivo* gating system by stably expressing human P-selectin on the target neuroblastoma cells. We engrafted 2 million GD2⁺ SK-N-BE(2) neuroblastoma cells in the flank of NSG mice with one cohort expressing P-selectin and another cohort P-selectin null to model *cis* synNotch CAR T cell killing (Fig. 6A and fig. S14). We hypothesized that α P-selectin synNotch GD2 CAR T cells would actuate in the TME of P-selectin⁺ tumors and cause target cell killing without trafficking and toxicity in the brain. Furthermore, we postulated that the synNotch CAR T cells would be in the “off state” in the absence of P-selectin and have no impact on tumor growth.

In P-selectin⁺ tumors, α hSCLP synNotch $\rightarrow$ GD2 CAR T cells led to a significant delay in tumor growth, relative to control mock T cell-treated mice (Fig. 6B). There was no significant change in tumor growth for synNotch-treated mice in the P-selectin-negative cohort (Fig. 6C). This suggests that there was no “leakage,” or non-specific CAR expression in the absence of the actuator ligand. While we found substantial infiltration of CD4⁺ and CD8⁺ T cells in constitutive GD2 CAR-treated mice, there was no detectable T cell infiltration in the α hSCLP synNotch $\rightarrow$ GD2 CAR T cell cohort (Fig. 6D). As previously observed, constitutive GD2 CAR-treated mice had to be euthanized because of rapid weight loss (>20%) (Fig. 6, E and F). Conversely, there was no observed weight loss in α hSCLP synNotch $\rightarrow$ GD2 CAR T cell-treated mice, suggesting a lack of systemic toxicity.

All MEAT cell-treated mice were ultimately euthanized because of tumor burden (>2000 mm³) and not because of weight loss or signs of neurological distress, demonstrating that P-selectin-gated

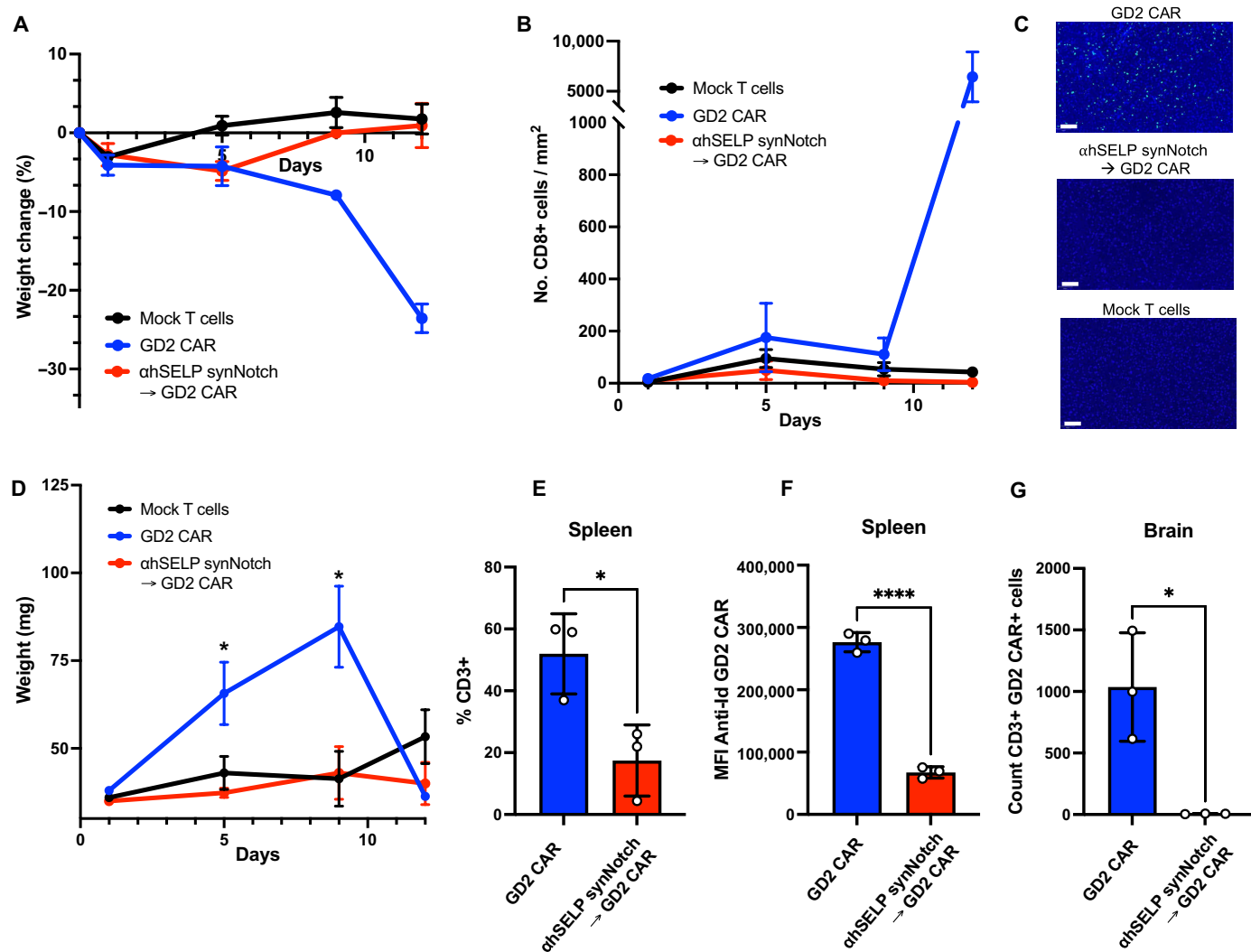


Fig. 5. MEAT cells are safe and well tolerated in mice. (A) Weight change in mice treated with 3×10^6 constitutive GD2 CART cells, ahSELN synNotch → GD2 CAR MEAT cells, or mock T cells (control). (B) Quantification of CD8+ cells in murine brain tissue on days 1, 5, 9, and 12 after T cell injection. (C) Representative IF staining of CD8+ cells on day 12 after T cell injection from (B) (scale bars, 100 μ m). (D) Spleen weight on days 1, 5, 9 and 12 after T cell injection. Ex vivo flow of CART cells at the end point (day 12) in the spleen (E and F) and brain (G). MFI, mean fluorescence intensity. Data are shown as means $\pm$ SD ($n = 3$ mice per group), and statistics were calculated using one-way ANOVA with Dunnett's post hoc test (D) or unpaired t test [(E) to (G)] where $*P \leq 0.05$ and $****P \leq 0.001$.

CAR T cells can ameliorate on-target, off-tumor neurotoxicity (Fig. 6, G and H). In mice bearing P-selectin-negative tumors, there was no survival benefit provided by any CAR T cell, because the synNotch-gated cells did not actuate to kill and the conventional GD2 CAR T cells killed the mice from toxicity more quickly than the control mice died from tumor burden. Similar trends were observed in a second solid tumor model where ahSELN synNotch → GD2 CAR T cells delayed GD2⁺ K562 tumor growth and prevented T cell brain infiltration (fig. S15).

α P-selectin synNotch CD19 CAR T cells improve antitumor efficacy in solid tumor models

Last, we sought to validate the MEAT cell system using a second CAR and solid tumor model. We engrafted NSG mice with SK-N-BE(2) neuroblastoma target cells stably expressing CD19, as previously described. Mice were treated with constitutive CD19 CAR T

cells, ahSELN synNotch → CD19 CAR T cells, or mock T cells, and tumor growth kinetics were monitored by a digital caliper. We observed similar trends as reported with the GD2 CAR, where P-selectin⁺ tumor growth was significantly delayed in mice treated with gated CAR T cells and there was no therapeutic efficacy against P-selectin⁻ tumors (Fig. 7, A and B). ahSELN synNotch → CD19 CAR T cells led to greater reductions in tumor sizes and a significant increase in overall survival in tumors that actuated the CAR (Fig. 7C). No improvement in survival or tumor reduction was observed in mice treated with MEAT cells if P-selectin was absent (Fig. 7D).

To further elucidate the mechanism of improved antitumor efficacy, we assessed T cell infiltration on days 7 and 14 after T cell engraftment in the spleen, lungs, and tumor by ex vivo flow cytometry (fig. S16A). We found a substantially higher population of CD3⁺ CD45⁺ T cells infiltrating the spleen and lungs 7 days after treatment in synNotch T cell-treated mice (Fig. 7H). Of the synNotch T

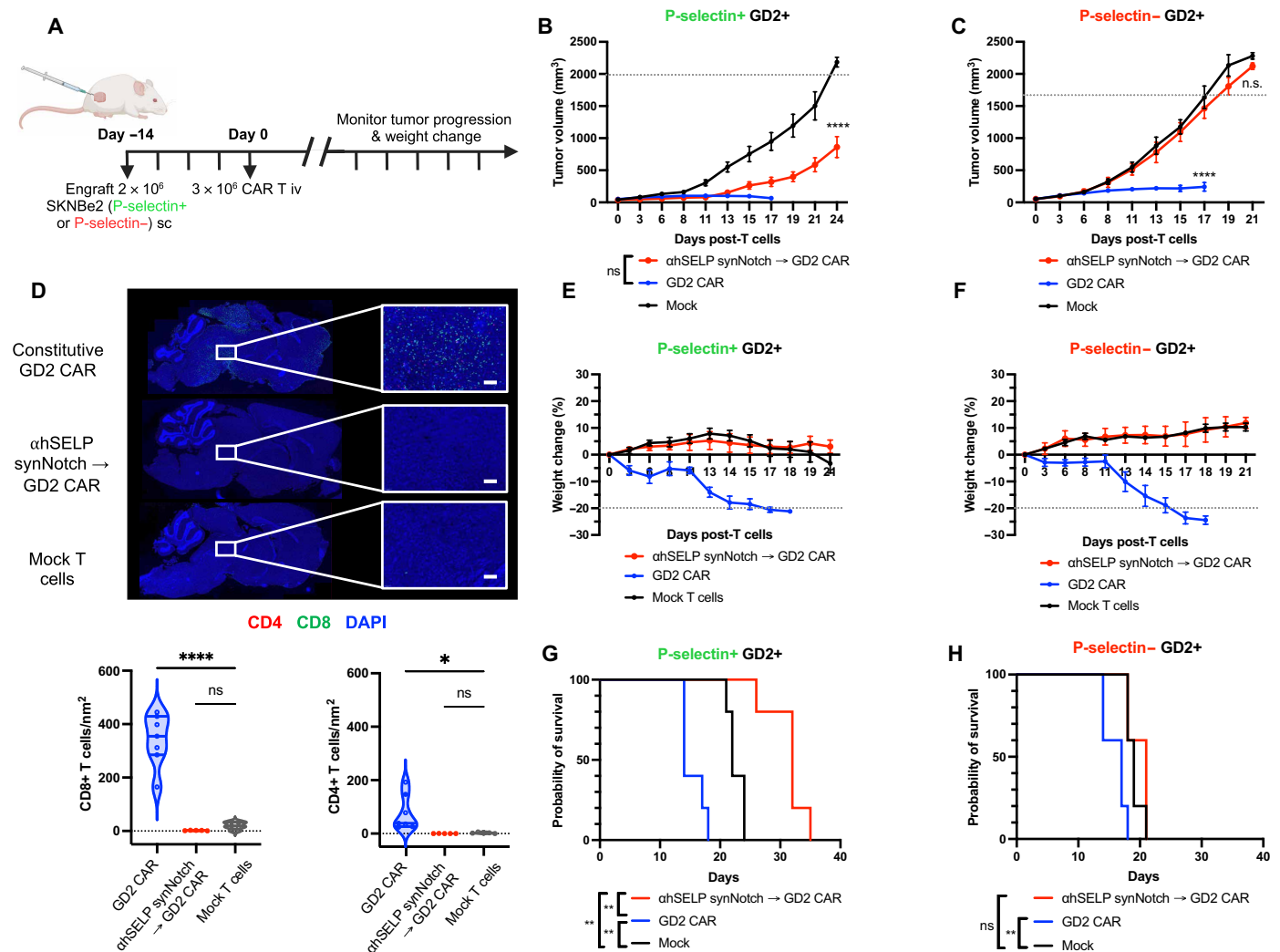


Fig. 6. P-selectin-gated GD2 CAR T cells ameliorate toxicity. (A) Experimental scheme to assess the efficacy of MEAT cells in a solid tumor model of neuroblastoma. Mice were treated with 3×10^6 constitutive GD2 CAR T cells, ahSELN synNotch → GD2 CAR MEAT cells, or control mock T cells 14 days after tumor inoculation. iv, intravenously; sc, subcutaneously. Tumor growth kinetics from SK-N-BE(2) tumor-bearing mice with (B) or without (C) P-selectin. (D) Top: Representative IF staining of CD8+ and CD4+ cells in murine brain at the survival end point (>20% weight loss or >2000-mm³ tumor size). Scale bars, 50 μ m. Bottom: Quantification of T cell infiltration. Weight change in SK-N-BE(2) tumor-bearing mice with (E) or without (F) P-selectin. Kaplan-Meier survival curves from SK-N-BE(2) tumor-bearing mice with (G) or without (H) P-selectin. Data are shown as means $\pm$ SEM, and statistics were calculated using unpaired *t* test [(B) and (C)], one-way ANOVA with Tukey's post hoc test (D), or Kaplan-Meier survival analysis [(G) to (H)] where ns > 0.05, * $P \leq 0.05$ and **** $P \leq 0.0001$. $n = 5$ mice per group.

cells detected in normal secondary lymphoid tissues, there was no detectable CAR expression, demonstrating synNotch T cells in the “off state” outside of the TME. We found that ahSELN synNotch → CD19 CAR T cells persisted longer and had significantly higher infiltration of T cells in the tumor relative to the constitutive CD19 CAR, which was below the limit of detection in the tumor by both flow cytometry and IF (Fig. 7E and fig. S16B). Similar trends in antitumor efficacy and tumor-infiltrating lymphocyte (TIL) quantification were observed with a second T cell donor (figs. S17 and S18).

Lastly, we established a mixed antigen model to probe *trans* synNotch CAR T cell killing, simulating an environment where P-selectin is present on normal antigen-negative cells and the CAR target is present on cancer cells (Fig. 6G). We engrafted 2 million SK-N-BE(2) neuroblastoma cells in the flank of NSG mice with 20%

P-selectin⁺ and 80% CD19⁺ (P-selectin/CD19). A separate cohort of control mice was engrafted with 20% P-selectin⁻ and 80% CD19⁺ (WT/CD19). Mice were dosed with either 5 million constitutive CD19 CAR T cells or ahSELN synNotch → CD19 CAR T cells, and tumor composition was verified by ex vivo flow cytometry 19 days after tumor engraftment. We found that the WT/CD19 mixed model maintained a 20/80% ratio, while the P-selectin/CD19 tumors decreased from 20% to 6% P-selectin⁺, possibly because of differential growth kinetics of target cells stably expressing P-selectin (fig. S19, A and B). We then assessed TIL infiltration 10 days after T cell treatment by flow cytometry and found that synNotch CAR T cells had significantly higher tumor infiltration, as observed in the *cis* killing model (Fig. 7H and fig. S19C). There was no significant increase in synNotch T cell infiltration in control WT/CD19 mice,

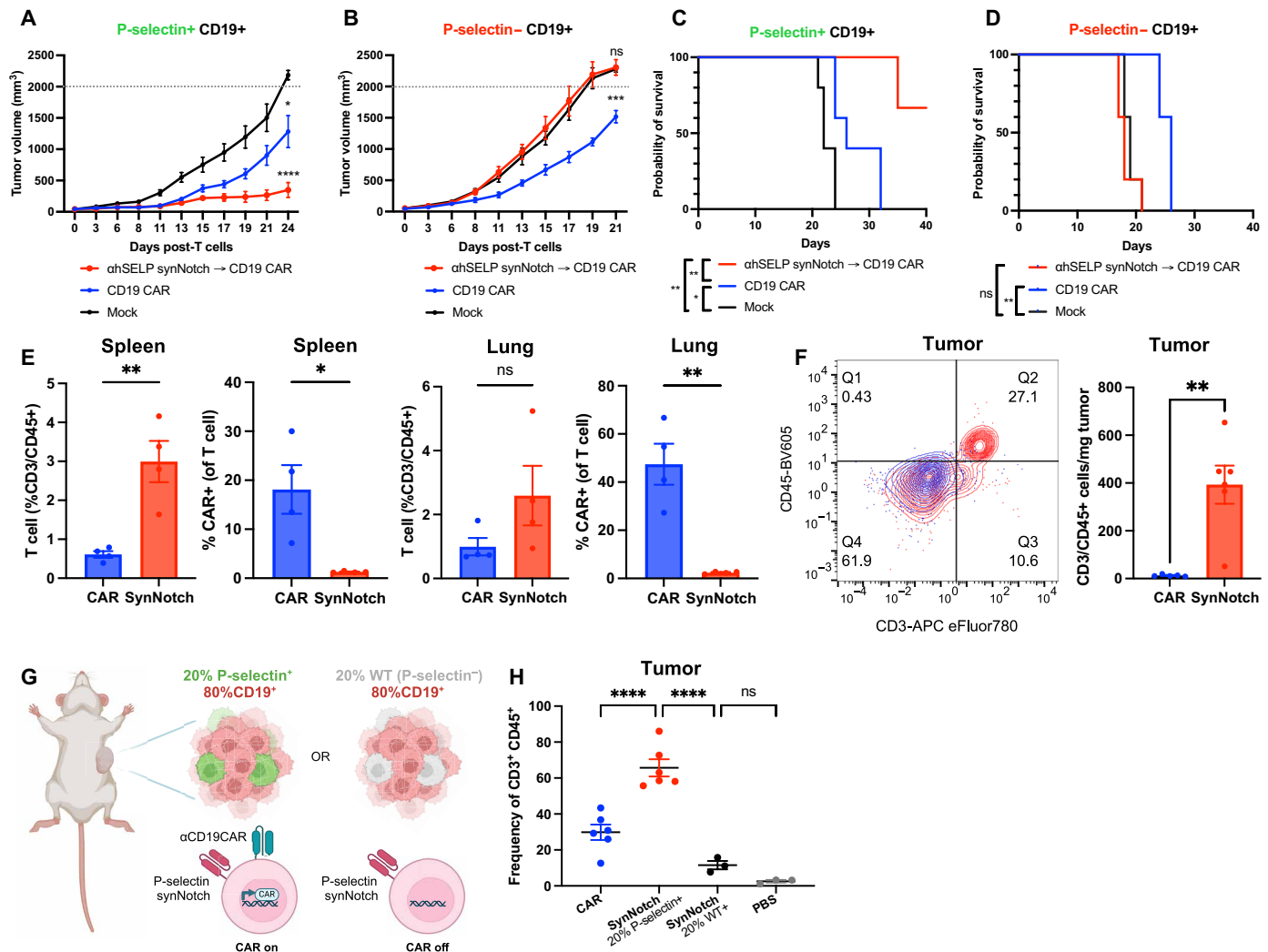


Fig. 7. P-selectin-gated CD19 CAR T cells improve T cell persistence and antitumor efficacy. Mice were treated with 3×10^6 constitutive CD19 CAR T cells, ahSELN synNotch → CD19 CAR MEAT cells, or control mock T cells 14 days after tumor inoculation. Tumor growth kinetics from SK-N-BE(2) tumor-bearing mice with (A) or without (B) P-selectin. Kaplan-Meier survival curves from SK-N-BE(2) tumor-bearing mice with (C) or without (D) P-selectin. Ex vivo flow cytometric analysis of (E) the spleen and lung and (F) tumor-infiltrating T cells. (G) Mixed antigen model to assess biodistribution of MEAT cells in a *trans* system. (H) Ex vivo flow cytometric analysis of tumor-infiltrating T cells 17 days after treatment in P-selectin⁺ tumors or P-selectin⁻ tumors. Data are shown as means $\pm$ SEM, and statistics were calculated using unpaired *t* test [(A), (B), (E), and (F)], Kaplan-Meier survival analysis [(C) and (D)], or one-way ANOVA with Tukey's post hoc test (H) where $ns > 0.05$, $*P \leq 0.05$, $**P \leq 0.01$, and $****P \leq 0.001$.

demonstrating specific biodistribution and expansion to P-selectin⁺ tissue. These data suggest that actuating via normal cells in the TME is a viable approach to gating toxic CAR T cells and that conditional CAR expression can improve cell fitness and function.

DISCUSSION

The MEAT cell platform established herein provides a potential means to overcome a major toxicity limitation for the use of CAR T cells in solid tumors through conditional CAR expression, which is activated only by specific cues in the TME. Here, we identified P-selectin as a highly tumor-specific antigen that can be sensed via synNotch circuits to then allow CAR T cell killing in combination with a second tumor antigen recognized by the CAR. MEAT cells gating the neurotoxic GD2 CAR circumvented lethal off-tumor reactivity

and showed complete amelioration of T cell infiltration in the brain while maintaining antitumor efficacy. Furthermore, MEAT cells gating the CD19 CAR showed enhanced control of tumor growth and a significantly higher persistence of T cells in the tumor 2 weeks after treatment, as compared to constitutive CD19 CAR T cells. We attribute these results to conditional CAR expression enabling (i) a less differentiated preinfusion product and (ii) a more metabolically fit cell that favors a naïve-like oxidative metabolism.

While gating mechanisms have been described for selective actuation of T cells by use of the synNotch signaling pathway, appropriate actuating signals within the TME are not well known and successful use of such systems to prevent toxicity while maintaining antitumor activity has not been described. This work constitutes the first application of a tumor vasculature ligand to actuate CAR T cells, the first comprehensive study of conditional CAR T cell metabolism

and differentiation status, the first demonstration of complete amelioration of affinity-enhanced GD2 CAR T cell brain infiltration, and the first evidence of substantial enhancement of antitumor efficacy using TME-gated CAR T cells.

P-selectin was chosen as the actuator target partly because of its high expression in a wide range of solid cancers, including ovarian, lung, breast, and neuroblastoma, with minimal expression in normal adjacent tissue. We and others have also reported minimal P-selectin expression in the midbrain, periventricular, hippocampal, and white matter regions in normal brain tissue, with only the choroid plexus region of the central nervous system expressing detectable levels (27, 37–40). In addition, targeting an actuator ligand present on normal cells, such as P-selectin on endothelial cells within the TME, minimizes the potential for antigen escape because of reduced plasticity and heterogeneity. If the actuator was on the neoplastic cell, escape via mutation or down-regulation is more likely. P-selectin was further selected for its association with T cell trafficking and migration to sites of inflammation, enabling CAR T cells to harness their endogenous trafficking modalities to achieve tumor homing. P-selectin can therefore act as a sole target for CAR T cell therapy or serve in combination with tumor-associated antigens in logic-gated circuits. In this way, the MEAT system is a modular platform that can be adapted to gate other cell therapies limited by lethal toxicity, such as the ROR1 and HER2 CARs or the MAGE-A3 and MART-1 T cell receptors (14, 15, 48–50).

Further investigation on the impact of T cell targeting to P-selectin is warranted, as it may enable improved homing and infiltration into solid tumors. We demonstrated that MEAT cells specifically actuate and kill in response to P-selectin and are highly sensitive and cytotoxic, even when P-selectin is expressed on a fraction (2.5%) of the total cell population. While only 3% of the total cells in samples from patients with neuroblastoma were P-selectin⁺, 74% of endothelial cells expressed P-selectin, supporting its utility as a TME actuator target. P-selectin has been shown to be up-regulated by ionizing radiation, which may enable tumors expressing low levels of ligand to induce target expression for MEAT cell targeting (27, 33). P-selectin can also be induced by pro-inflammatory cytokines, providing a potential positive feedback loop of T cell homing, CAR expression, target cell death, and further up-regulation of P-selectin via CAR T cell-secreted cytokines (33, 51). Systemic inflammation caused by activation of CAR T cells within the TME should be closely monitored for potential up-regulation of P-selectin in normal tissue, which could act as a potential mechanism of off-tumor toxicity.

We first used the MEAT system to gate the affinity-enhanced GD2 CAR, as it was shown to have better tumor growth control over the WT CAR, although limited by neurotoxicity. Clinical trials have established a moderate benefit (33% of patients are complete responders) of GD2 CAR T cell treatment in neuroblastoma patients, although the less potent WT scFv was used in combination with an inducible Cas9 kill switch and dual CD28 and 4-1BBz costimulatory domains (13). Clinical use of the E101K GD2 CAR via MEAT cell gating may increase the fraction of responders while preventing lethal toxicity.

A recent report describes the selective actuation and killing of GD2 CAR T cells gated against tissue expressing B7H3, an immune checkpoint molecule expressed on solid tumors (21). Although mice did not develop signs of neurotoxicity, there was significant infiltration of T cells in the brain after treatment, posing a risk to patients should the gated system become leaky over time. Furthermore,

targeting two antigens on the cancer cell itself risks loss of either signal or escape by cancer cells. As such, patients may benefit from GD2 CAR T cells gated against the normal TME ligand P-selectin as they maintain the cytotoxic ability of the higher affinity CAR while completely abolishing T cell infiltration in the brain and minimizing risk to patients.

To demonstrate the modularity of the MEAT cell approach, we developed a second model system using a P-selectin-gated CD19 CAR. We found that MEAT cells outperformed constitutive CAR T cells via improved antitumor efficacy and prolonged TIL persistence. Analysis of the preinfusion CAR T cell products showed that MEAT cells expressed lower levels of exhaustion markers and maintained a more naïve-like phenotype. It has been demonstrated that ex vivo culture of CAR T cells promotes cell differentiation and that decreased stem cell memory fractions in the preinfusion product correlates with diminished persistence and cytotoxicity in vivo (23, 52–54). We hypothesize that synNotch gating offers a protective advantage to T cells by reducing tonic signaling and enabling a more fit preinfusion product. Furthermore, after eliminating target cells, or in the absence of the actuator ligand, synNotch CAR T cells reinternalize the CAR, preventing further exhaustion and prolonging cytotoxic function. Recent work has shown that CAR T cells treated with inhibitors, such as dasatinib, can modulate tonic signaling and improve memory formation, highlighting the need for intermittent rest of CAR T cells through modulation of CAR expression (55). In addition to enabling transient rest and reducing tonic signaling during manufacture, synNotch cells may also exhibit reduced tonic signaling because of the use of the weaker TRE promoter, as compared to the strong phosphoglycerate kinase (PGK) promoter used in the constitutive CAR (56). While incorporation of a weaker promoter to constitutive CARs may reduce tonic signaling, the CAR expression profile has been reported to be highly variable, leading to a less uniform T cell therapeutic (24).

T cell metabolism has been shown to be a key regulator of cell expansion and differentiation, and control of cellular metabolism during manufacture may enable a more fit cellular therapy. Analysis of OCR and ECAR via mitochondrial stress tests suggests that unstimulated MEAT cells have a lower basal level of glycolysis, demonstrating a more naïve, resting phenotype. Similar trends in oxygen consumption are observed in chronically stimulated effector cells, demonstrating favorable bioenergetics even after serial killing of target cells. These data support the observation that synNotch cells maintain a more naïve-like state and have a higher expansion potential compared to constitutive CAR T cells. Other groups have attempted to reprogram T cell metabolism during manufacture to maintain a less differentiated state using glycolysis inhibitors or by promoting mitochondrial function (43). Here, we show that simply enabling temporal control of CAR expression enables metabolic flexibility in synNotch cells, promoting a less differentiated preinfusion product and a more cytotoxic therapy after chronic stimulation. Further investigation is warranted to better understand the ability of conditional CAR expression to promote metabolic flexibility and the potential implications of this on T cell biology.

In conclusion, we establish P-selectin as a potential target within the TME for combinatorial synNotch circuits. The MEAT cell system offers superior antitumor efficacy in solid tumor models without associated lethal toxicities. We attribute improved efficacy to a less differentiated preinfusion product, higher metabolic fitness, and sustained persistence of TILs in vivo. Together, this

work develops a first-in-class approach to efficiently restrict expression of CARs to the TME to improve CAR T cell fitness, eliminate off-tumor toxicity, and expand the therapeutic window of CAR T cells in the treatment of solid tumors.

MATERIALS AND METHODS

Study design

The aim of this study was to evaluate P-selectin as a therapeutic actuator target in the TME to ameliorate off-tumor toxicities and to improve cell fitness of CAR T cells against solid tumor targets. We generated synNotch against human P-selectin to gate either the neurotoxic GD2 CAR or the CD19 CAR and assessed the functionalities of these MEAT cells against cancer cell lines *in vitro*. We then investigated exhaustion marker expression, memory phenotype, and cellular bioenergetics to evaluate the ability of gated systems to improve cell fitness. MEAT cells were then assessed for safety, specificity, and antitumor efficacy in multiple solid tumor xenograft models, including a mixed antigen *trans* model. Sample sizes for mouse experiments were selected on the basis of the previously published literature and experience in the lab, including a minimum of four mice per group. Tumor size was measured by a digital caliper or BLI, and mice were randomized to each experimental group before treatment administration. The number of experimental and biological replicates is indicated in the figure legends. Rules for stopping data collection were determined before the start of the experiment. Investigators were not blinded to the experiments, except for animal facility staff who administer health checks if mouse euthanasia is required.

Cloning and generation of lentiviral vectors and cell lines

The GD2 AkaLuc construct was generated by cloning the AkaLuc sequence downstream of the CAR using the pLenti-PGK-Venus-AkaLuc (neo) plasmid (a gift from R. Friedel, Addgene plasmid no. 124701) (57). MEAT cell systems were built by using the two-vector Gal4 synNotch receptor technology. The sensor and response elements were cloned into a single pHR lentiviral vector by Gibson assembly using the pHR_PGK_LaG16_2_synNotch_Gal4VP64 and pHR_Gal4UAS_tBFP_PGK_mCherry plasmids (a gift from W. Lim, Addgene plasmid nos. 85421 and 79130) (17, 18). The anti-human P-selectin scFv (derived from crizanlizumab) sequence was synthesized by Genscript. The affinity-enhanced E101K anti-GD2 CAR vector was provided by the Milone group from the University of Pennsylvania and was cloned into (i) a pHR lentiviral vector (constitutive GD2 CAR) or (ii) the response element of the MEAT system (ahSELP synNotch → GD2 CAR) by Gibson assembly. CD19 CAR and ahSELP synNotch → CD19 CAR vectors were generated using the same approach or directly synthesized by Genscript.

Primary human T cell isolation and culture

Peripheral blood mononuclear cells were isolated from healthy human donors using lymphocyte separation media. Blood was taken after written informed consent on MSK institutional review board-approved protocols. Red blood cells were lysed with ACK lysis buffer, and T cells were stimulated with anti-CD3 antibody (50 ng/ml; OKT-3, MACS) and IL-2 (100 IU/ml) in complete RPMI media.

Lentiviral transduction of T cells and cell lines

Lenti-X 293T (Lenti-X) cells were plated in 10-cm poly-D-lysine-coated tissue culture-treated plates. After 24 hours, Lenti-X cells

were transfected with equimolar concentrations of pHR, PAX2, and MD2.G plasmids in RPMI using Lipofectamine and P3000. Media were replaced 24 hours after transfection. Virus was harvested 48 and 72 hours after transfection and purified through a 0.45- μ m filter. Cell lines [Jurkat, SK-N-BE(2), T98G, Lenti-X, and K562] were ordered from American Type Culture Collection, transduced with viral supernatant supplemented with polybrene (4 μ g/ml), and passaged after 72 hours. T cells were transduced 48 hours after stimulation in six-well nontreated tissue culture plates coated with retronectin, and viral supernatant was added with polybrene (4 μ g/ml). Cells were spun at 2000g for 1 hour. T cells were spinoculated again at 72 hours after stimulation.

Flow cytometry

Transduction efficiency was verified by flow cytometry using an AF647- α Myc tag antibody (synNotch receptor, clone 71D10, Cell Signaling), APC- α GD2 CAR (clone 1A7, Creative Biolabs), or AF647- α CD19 CAR (clone 19E3, generated at MSK Antibody and Bioresource Core Facility). Tumor tissue was processed for flow cytometry using the human tissue dissociation kit (Miltenyi) and gentleMACS Octo Dissociator for 1 hour on gentle setting. Red blood cells were lysed using ACK buffer for 5 min at room temperature and the lymphocyte fraction enriched using a 40/80 Percoll gradient. All samples were washed and stained in fluorescence-activated cell sorting buffer [2% fetal bovine serum and 0.1% azide in phosphate-buffered saline (PBS)] at 4°C. Data were collected using a MACsQuant16 flow cytometer (Miltenyi) or a CytoFLEX LX (Beckman Coulter). FlowJo software (version 10.9.0) was used for all data analyses.

Actuation assays

MEAT cell actuation (induction of surface CAR) was assessed via coculture with P-selectin. Briefly, 1 μ g of recombinant human P-selectin (R&D Systems) was coated on a nontreated 96-well plate at 4°C overnight. MEAT cells were then seeded at 50,000 cells/ml in wells containing coated P-selectin, soluble P-selectin, or PBS control. Cells were harvested after 24 hours to measure surface CAR expression by flow cytometry.

Cytotoxicity assays

The cytotoxicity of constitutive CAR and synNotch CAR T cells was determined using an Incucyte S3 Live-Cell Analysis system (Sartorius). Target cells [SK-N-BE(2) or T98G] expressing mCherry and actuator cells [Lenti-X stably expressing P-selectin fused to GFP (green fluorescent protein)] were cocultured in triplicate with effector cells at indicated E:A:T ratios using clear-bottom, white 96-well assay plates (Corning 3903) in a total volume of 200 μ l. Images were acquired every 2 hours, and mean fluorescence intensity was calculated using Incucyte Live-Cell Analysis software and normalized to target cells alone, unless stated otherwise. Data were collected in three technical replicates and summarized as means $\pm$ SD. All cytotoxicity assays were repeated with at least two additional T cell donors for a total of three biological replicates, and all repeats support the representative data shown in the results.

Cytokine profile analysis

Effector cells were co-incubated with SK-N-BE(2) target cells expressing both the actuator and antigen (P-selectin⁺ and CD19⁺) or only one present (P-selectin⁺ or CD19⁺) at 2:1 E:T. Conditioned media were collected 72 hours after incubation and spun down at

500g for 10 min. The supernatant was filtered through a 0.45- μ m filter and snap frozen at -80°C until all donors were collected for analysis. Cytokine composition was then profiled on a Luminex 200 using the ProcartaPlex Human Cytokine Panel 1B, 25plex (Thermo Fisher Scientific). Data were collected in technical duplicates for a total of four biological replicates and summarized as means $\pm$ SD.

T cell activation assays

CAR T cells were acutely activated with either WT SK-N-BE(2) cells (P-selectin⁻) or SK-N-BE(2) cells overexpressing P-selectin via reverse time course. Briefly, 10,000 target cells were seeded in a 96-well plate the day before activation. T cells were then seeded in fresh F12:minimum essential medium complete media at an E:T of 2:1 at 72, 48, 24, and 0 hours. CD69 was measured at the end of the time course via flow cytometry. Data were collected in three technical replicates and summarized as means $\pm$ SD. All activation assays were repeated with at least two additional T cell donors for a total of three biological replicates, and all repeats support the representative data shown in the results.

CAR T cells were chronically activated for 15 days by coculture with 20% actuator (P-selectin⁺) and 80% target (CD19⁺) cells (E:T of 2:1). T cells were harvested, counted, and reseeded in fresh complete RPMI in 10-cm tissue culture dishes with fresh actuator and target cells on days 4, 8, and 12 for a total of four stimulations. T cells were harvested on day 15 for analysis of exhaustion and memory phenotype by flow cytometry, cytotoxicity by live-cell killing, and metabolic changes by Seahorse mitochondrial stress test. Killing data were collected in technical triplicates for a total of two biological replicates and summarized as means $\pm$ SD. Exhaustion marker analysis is shown for three biological replicates and summarized as means $\pm$ SD.

Seahorse metabolic analysis

Mitochondrial function was assessed using an XFe96 extracellular flux analyzer (Seahorse Bioscience). The sensor cartridge was hydrated in calibrant solution overnight in a 37°C non- CO_2 incubator. Effector cells were cocultured with 20% P-selectin⁺ actuator cells and 80% CD19⁺ target cells for 48 hours at a 2:1 ratio to induce *trans* actuation and killing. A 96-well microplate was coated with poly-D-lysine for 30 min at room temperature and washed twice with PBS. T cells were isolated from coculture via positive selection using CD45 nanobeads (Biolegend) and seeded at 0.2×10^6 cells per well in XF assay medium (nonbuffered RPMI supplemented with 10 mM glucose, 1 mM sodium pyruvate, and 2 mM L-glutamine). The microplate was centrifuged at 300g for 1 min and incubated in a 37°C non- CO_2 incubator for 60 min. OCR and ECAR were measured after serial treatment of 1 μM oligomycin, 1.5 μM FCCP, and 0.5 μM rotenone/antimycin A. Data were collected in at least six technical replicates and summarized as means $\pm$ SD. All Seahorse assays were repeated with at least one additional T cell donor for a total of two biological replicates.

Immunofluorescence staining

Tissue microarray slides were purchased from the vendor Tissue Array (cat. no. NB642c neuroblastoma array and cat. no. GLN241a normal cerebrum array). Murine tissue (brain and subcutaneous tumor) was harvested and fixed in 4% paraformaldehyde at 4°C for 18 to 24 hours. Tissue was then dehydrated in 70% ethanol for 48 hours and embedded in paraffin. Sections (5 μM) were sliced and mounted

on microscope slides. Tissue sections were then washed with $1\times$ PBS, and antigen retrieval was performed with CC1 buffer (Ventana Medical Systems). Sections were blocked for 30 min with Background Buster solution (Innovex no. NB306).

The IF staining of CD4, CD8, CD45, CD31, and P-selectin was performed at the Molecular Cytology Core Facility (MSKCC). Briefly, slides were stained using a Discovery XT processor or Ultra processor (Ventana Medical Systems-Roche). Tissue sections were blocked for 30 min in Background Blocking solution after 32 min of heating and standard retrieval via Cell Conditioning 1 (Ventana no. 950-500). A rabbit monoclonal anti-CD31 (Abcam no. ab182981) was used at 0.16 $\mu\text{g}/\text{ml}$ and incubated for 5 hours. A biotinylated goat anti-rabbit IgG secondary antibody (Vector Labs no. PK6101) was then incubated at 5.75 $\mu\text{g}/\text{ml}$. Blocker D, streptavidin-horseradish peroxidase, and trichostatin A (Life Tech no. B40932) were prepared and applied according to the manufacturer's instruction at 1:150 for 16 min. The procedure was repeated using a rabbit polyclonal anti-P-selectin antibody (LSBio no. LSB3578) at 2.5 $\mu\text{g}/\text{ml}$ for 4 hours. A biotinylated goat anti-rabbit IgG secondary antibody was then incubated at 5.75 $\mu\text{g}/\text{ml}$. All slides were counterstained with DAPI (4',6-diamidino-2-phenylindole; 5 $\mu\text{g}/\text{ml}$) for 5 min at room temperature, mounted with anti-fade mounting medium, and sealed with a coverslip. The same procedure was repeated for staining of CD4 (Roche, Ventana no. 790-4423, 1:20 dilution), CD8 (Roche, Ventana no. 790-4460, 1:20 dilution), and CD45 (Dako no. M0701, mouse monoclonal, 1.25 $\mu\text{g}/\text{ml}$).

High-resolution scans of stained slides were then acquired by the MSKCC using a confocal microscope scanner. Images were quantified using QuPath software via cell detection and object classification for each channel of interest. Tissue microarrays of human neuroblastoma samples were provided as duplicate cores per patient ($n = 27$ unique patients) and quantified as individual data points ($n = 54$ cores) in Fig. 2B.

In vivo studies

All animal experiments were performed in compliance with MSKCC IACUC-approved protocol 96-11-044. NOD.Cg-Prkdc^{scid} Il2rg^{tm1Wjl}/SzJ (NSG) mice from the Jackson Laboratory (strain no. 005557) were used for in vivo studies. Mice were euthanized at tumor burden $>2000 \text{ mm}^3$ or weight loss $>20\%$ as per the IACUC protocol.

SK-N-BE(2) solid tumor model

NSG mice were engrafted subcutaneously with 2×10^6 SK-N-BE(2) target cells (either stably expressing human P-selectin or WT P-selectin null) using a 1:1 mixture of Matrigel and RPMI on day 0. Tumor burden was measured on day 14 after tumor engraftment by a digital caliper, and mice were randomized into treatment groups. Mice were injected intravenously with 3×10^6 MEAT cells (CD19 or GD2), constitutive CAR T cells (CD19 or GD2), or mock T cells ($n = 5$ mice per group). Tumor burden was monitored by a digital caliper, and the weight and signs of neurological distress were monitored through the end point. Efficacy data were repeated with one additional T cell donor and shown in the Supplementary Materials.

For the mixed antigen model, 2×10^6 SK-N-BE(2) target cells were engrafted in the flank of NSG mice at a ratio of 20% P-selectin⁺ to 80% CD19⁺ (P-selectin/CD19). Mice were dosed with either 5 million constitutive CD19 CAR T cells or ahSELP synNotch $\rightarrow$ CD19 CAR T cells ($n = 6$ mice per group) 14 days after tumor engraftment. A separate control cohort was established with 20% P-selectin⁻ and 80% CD19⁺ target cells (WT/CD19) and treated with ahSELP synNotch $\rightarrow$ CD19

CAR T cells or PBS control ($n = 3$ mice per group). These data were only collected with one T cell donor.

K562 solid tumor model

NSG mice were engrafted subcutaneously with 1×10^6 K562 target cells stably expressing AkaLuc, GD2, and human P-selectin using a 1:1 mixture of Matrigel and RPMI on day 0. Tumor burden was measured on day 2 after tumor engraftment via BLI using a Xenogen IVIS Spectrum, and mice were randomized into treatment groups. GD2 CAR MEAT cells (6×10^6 , 3×10^6 , or 1.5×10^6), constitutive GD2 CAR (1.5×10^6), or control mock T cells (6×10^6) were injected intravenously 3 days after tumor engraftment. Tumor burden was monitored by BLI, and the weight and signs of neurological distress were monitored through the end point. Living image software was used for all BLI analyses.

Statistical analysis

Statistics were calculated using paired or unpaired t test and two-way analysis of variance (ANOVA) with appropriate post hoc test as stated in the figure legends. Survival plots of solid tumor models were analyzed using the Mantel-Cox log-rank test. Statistical significance was indicated accordingly where $ns > 0.05$, $*P \leq 0.05$, $**P \leq 0.01$, $***P \leq 0.001$, and $****P \leq 0.001$. All statistical analyses were performed using GraphPad Prism version 9.1.0. No statistical methods were used to predetermine sample size. For all technical replicates reported, measurements were taken from distinct samples.

Cancer cell lines

Jurkat, K562, T98G, and SK-N-BE(2) cells were ordered from American Type Culture Collection. Lenti-X 293T viral producer cells were ordered from Takara Bio.

Supplementary Materials

This PDF file includes:

Supplementary Text
Figs. S1 to S19

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